

# HearMeOut: Speech-Based Mood Journaling Interface

## Hi-Fi Prototype Design

CS 435 - Human Computer Interaction

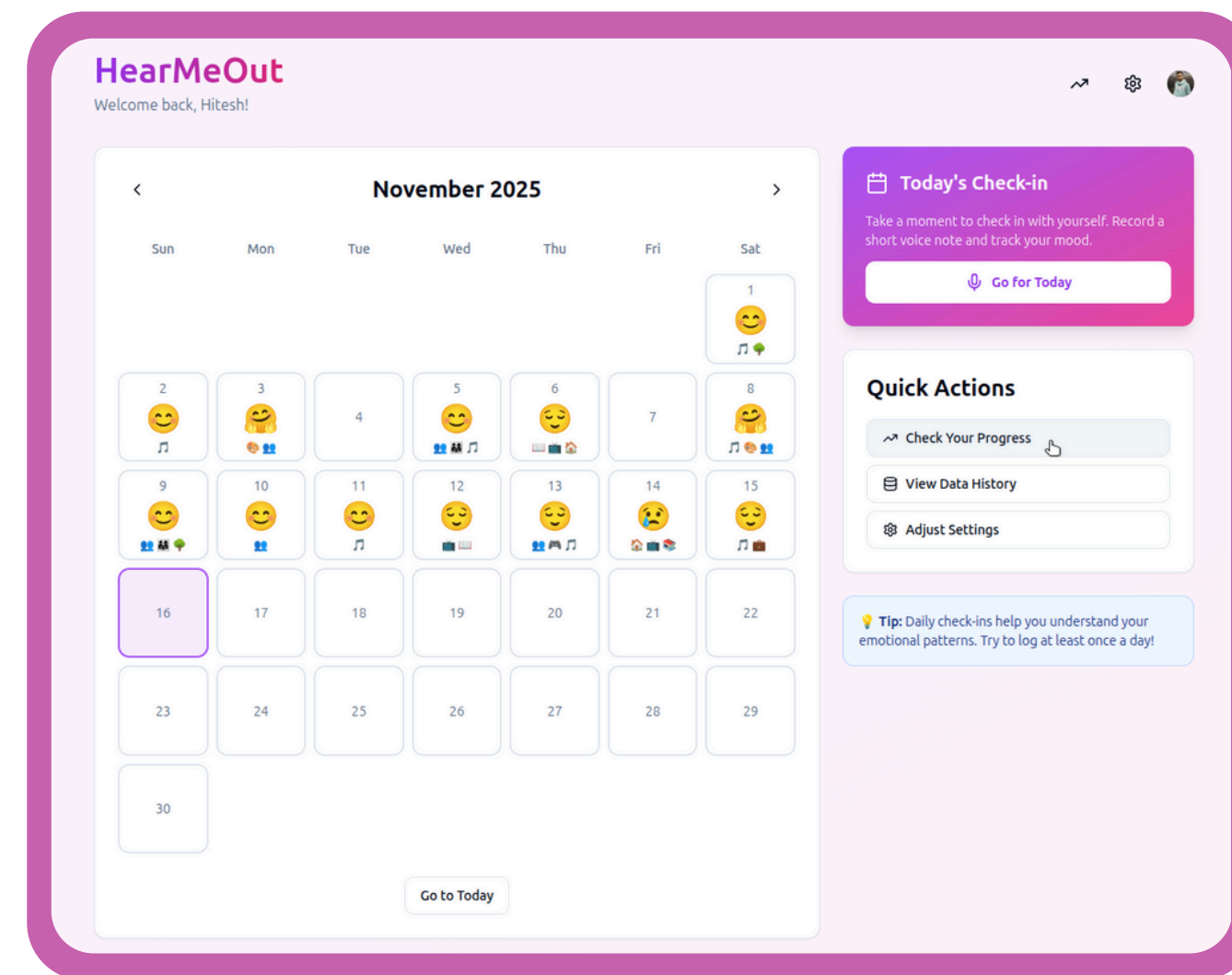
### TEAM - 6

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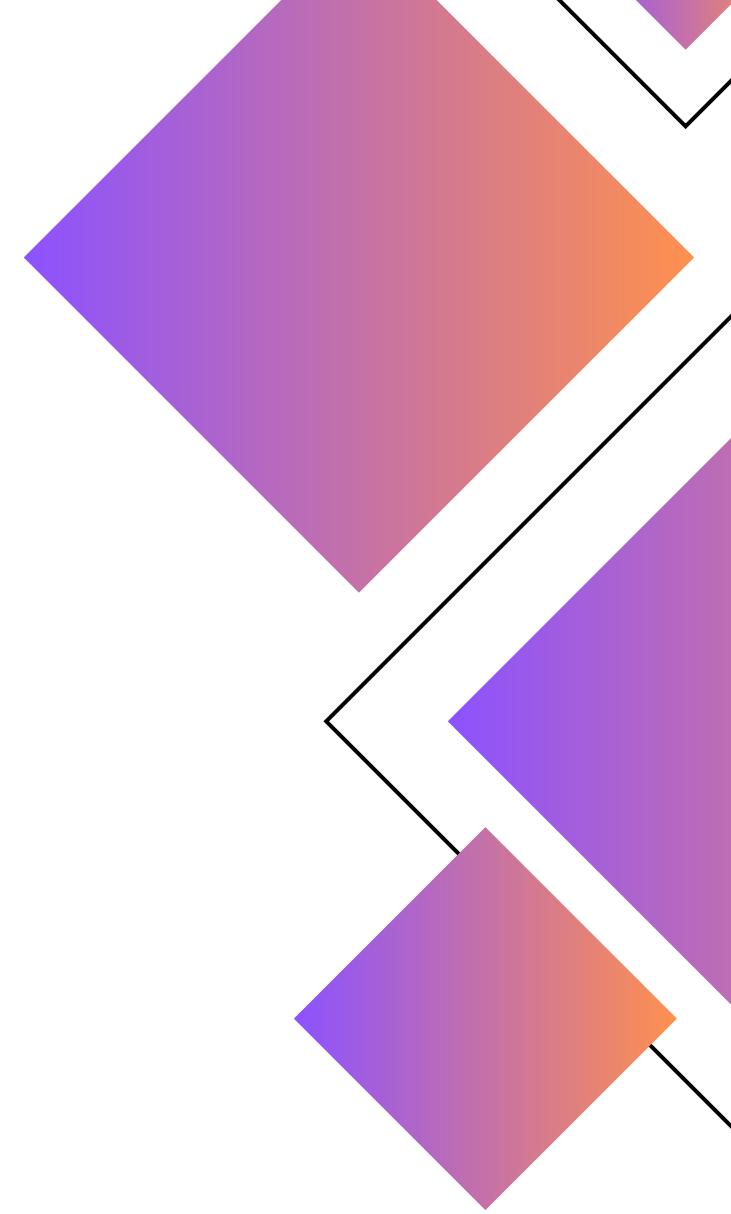
# **Problem & Motivation**

## **Why this project?**

- Student mental health issues are rising, but many hesitate to seek help due to stigma, fear of judgment, and limited access to counselors.
- Traditional assessments (self-reports, clinical interviews) miss gradual emotional decline.

## **Goal:**

Build a non-clinical, daily-use, speech-based mood tracking tool that feels natural, private, and unintimidating.



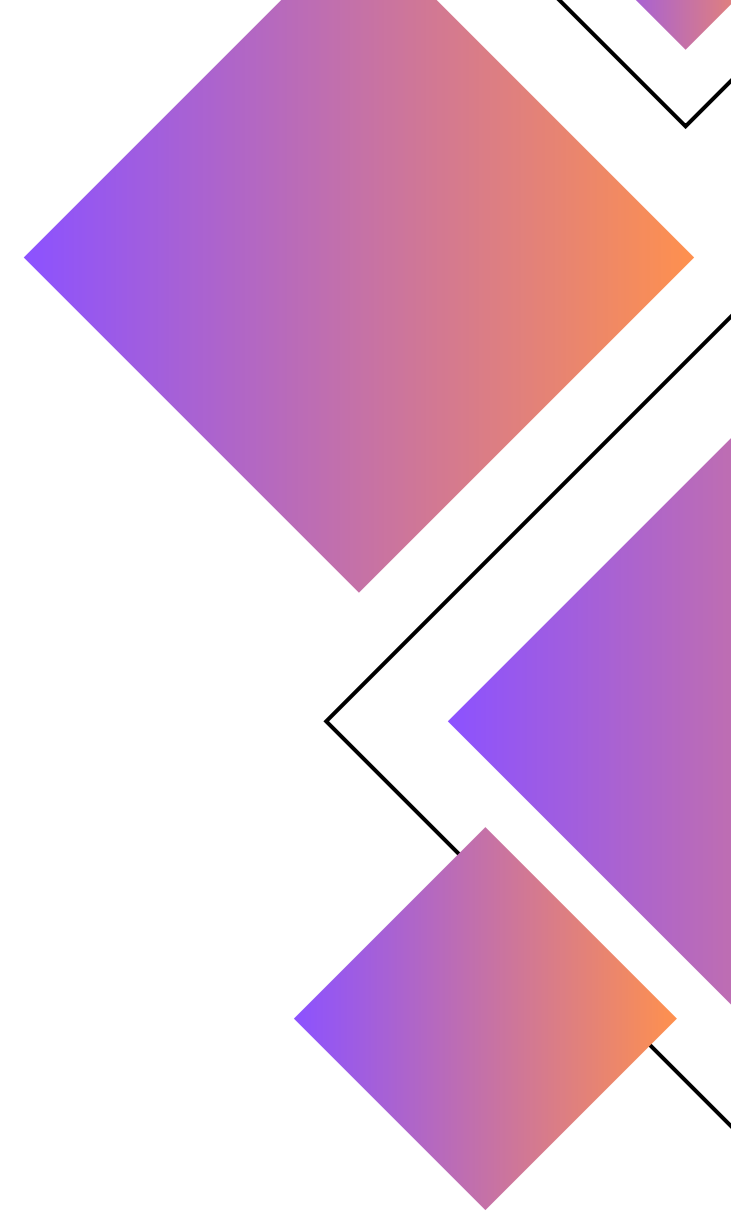
# Conceptual Model

## ***Primary Metaphor: Daily Mood Journaling***

- **NOT** a “Mental Health Screening” → A gentle *Journaling Companion*.
- Replaces Clinical Labels with Emojis and Reflections.
- Reduces Stigma and creates a Low-Pressure emotional check-in ritual.
- **Language Choices:**
  - “*Record today’s mood*” (not “Complete Assessment”)
  - “*Mood Journal*” (not “Depression Screening Tool”)
  - “*Feeling patterns*” (not “Symptom Severity”)
  - “*Suggested Emojis*” (not “AI Diagnosis”)

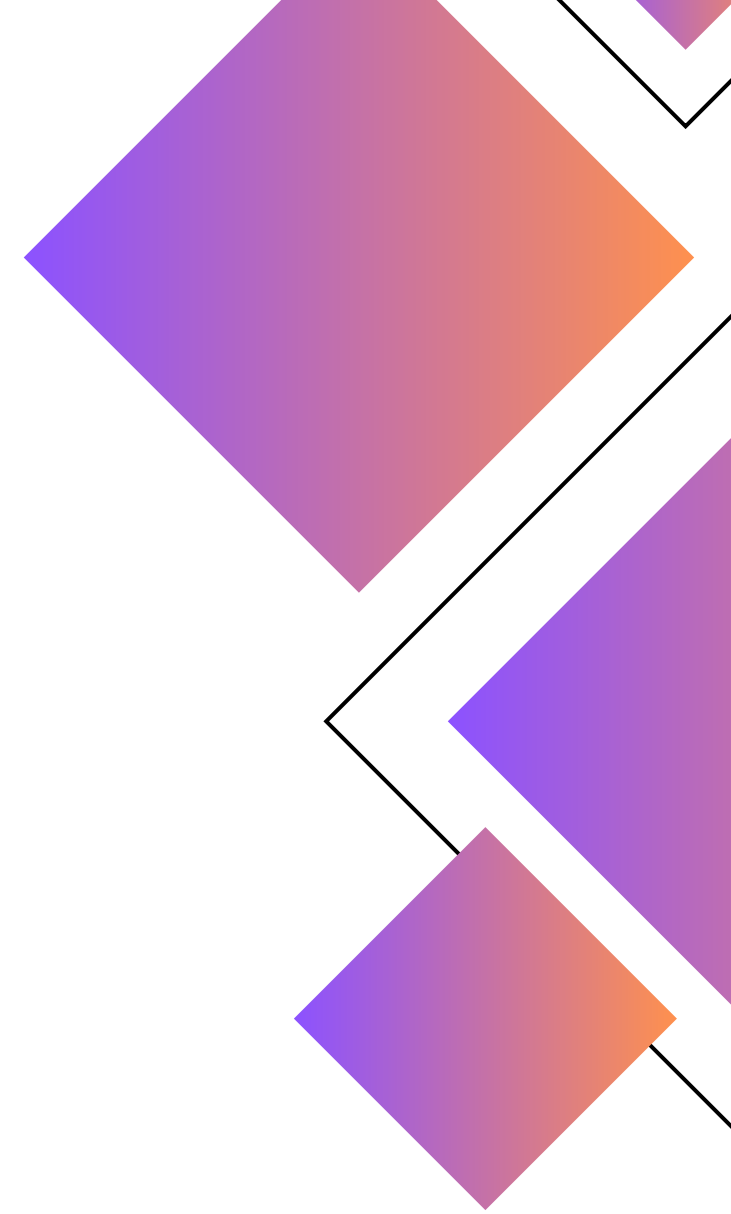
## **Mental Model:**

- AI acts as an **Assistant**, not an Authority.
- It suggests moods based on Speech, but the user always chooses the final emotion.
- Transparent confidence scores maintain trust without making the AI feel diagnostic.

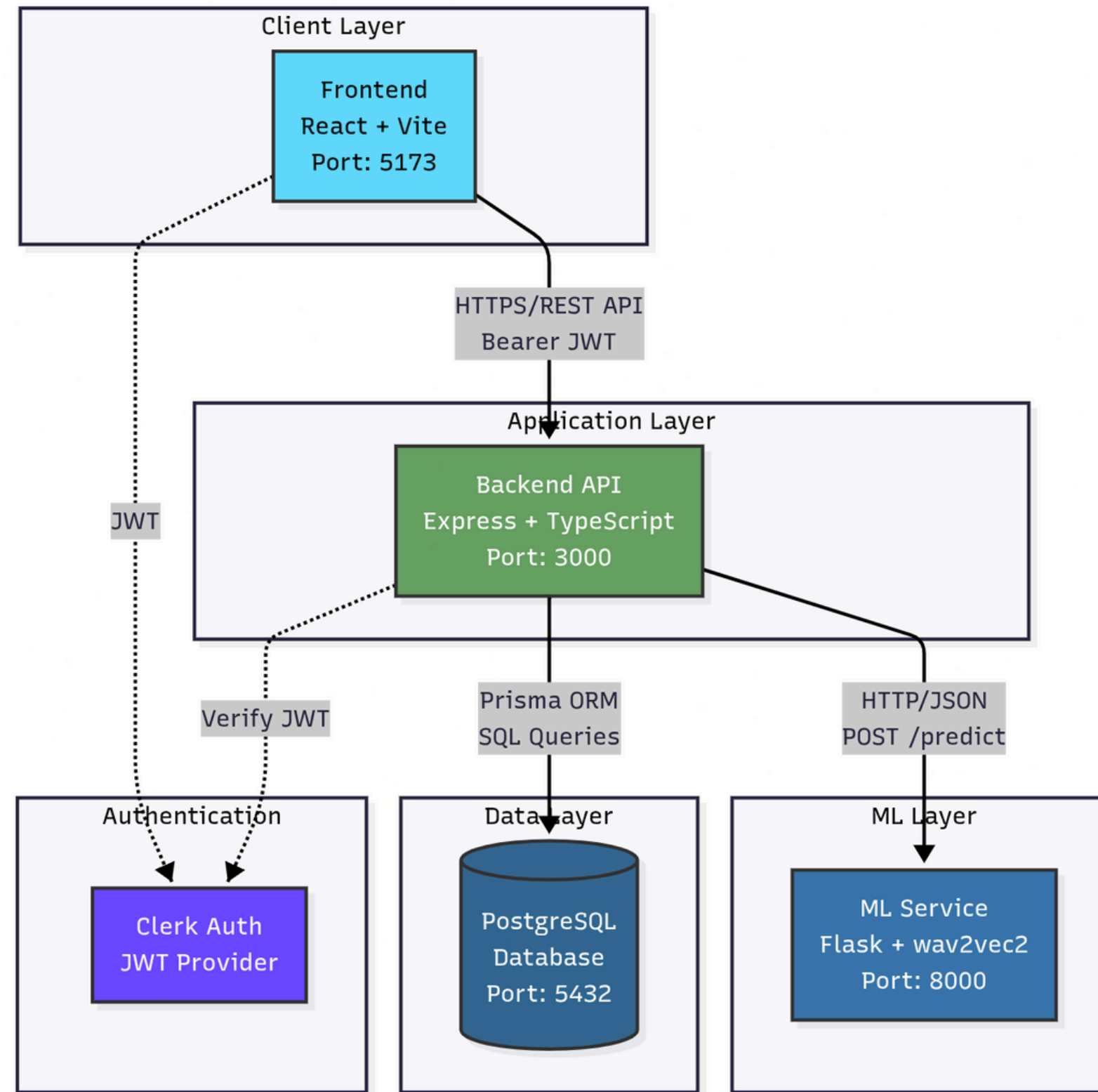


# Design Space: The 6 W's

1. **Who:** Students who avoid Clinical Tools or care deeply about Privacy.
2. **What:** Emotion recognition from Voice + Calendar-based Mood Tracking + Pattern Visualization.
3. **When:** Daily evening check-ins, Optional and Flexible.
4. **Where:** Private environments (dorms, rooms, quiet spaces).
5. **Why:** Self-awareness, Pattern Recognition, Early signals of Distress.
6. **How:** Speak → AI analyzes → User Picks Emoji → Calendar + Analytics Update.

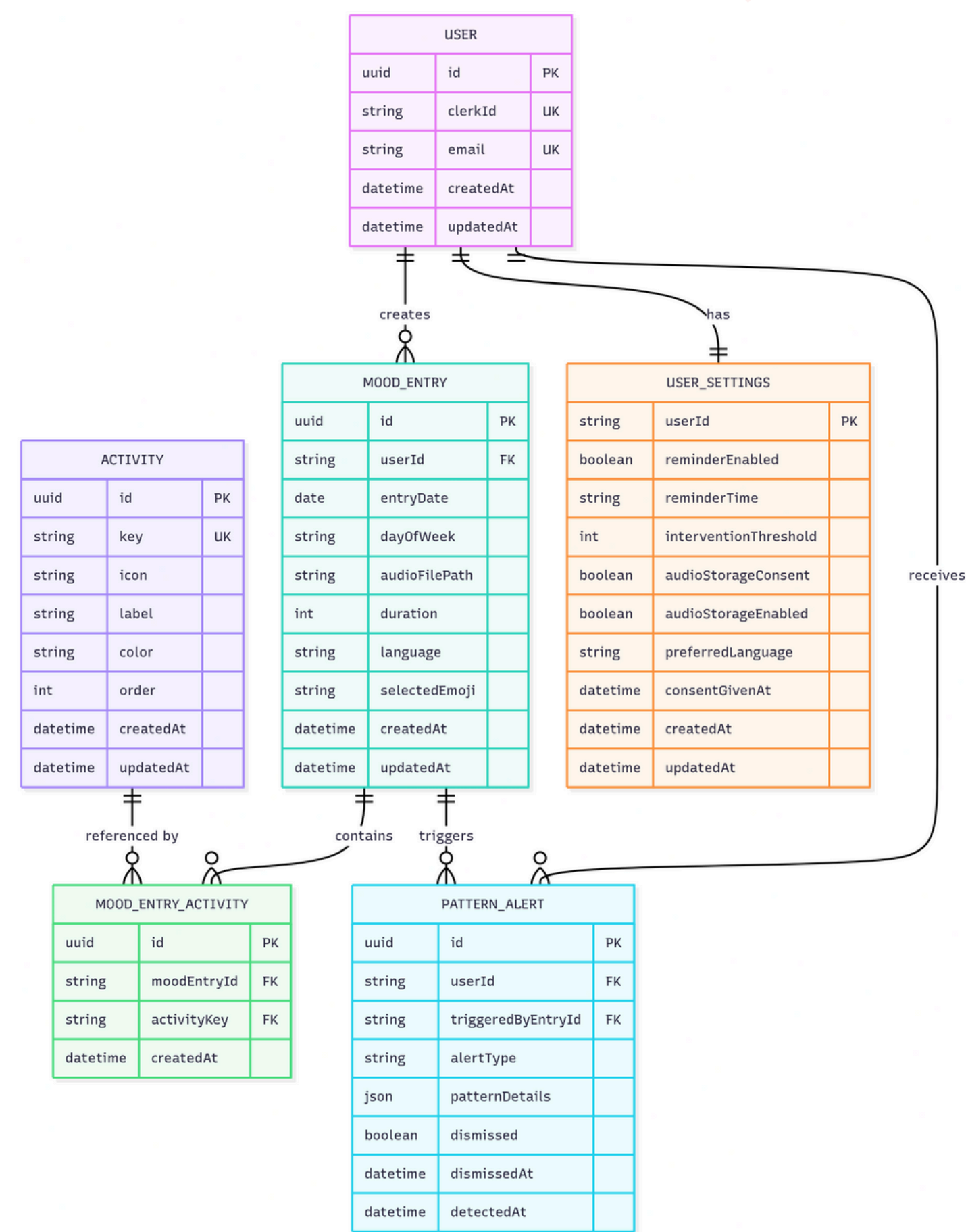


# System Architecture

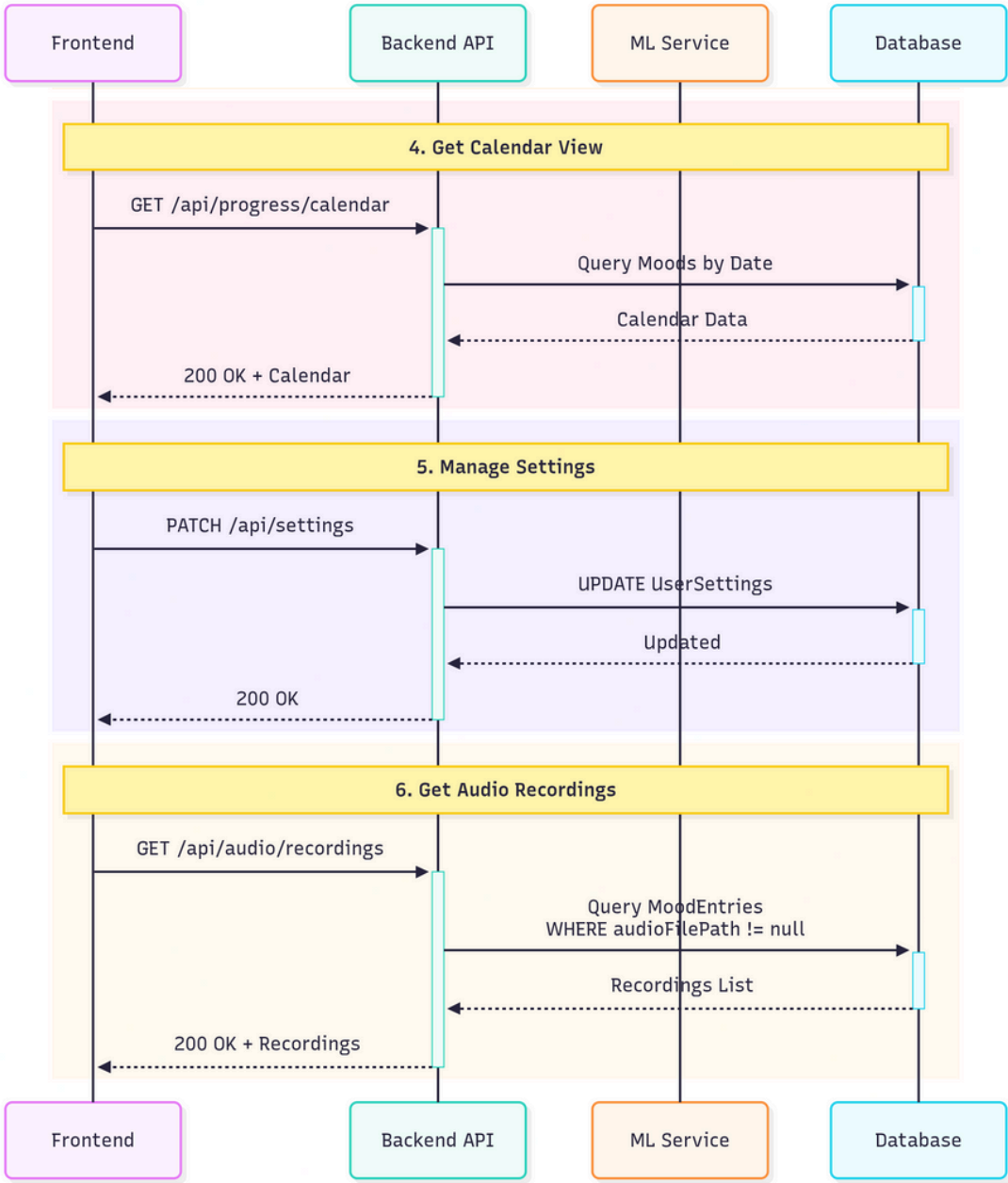
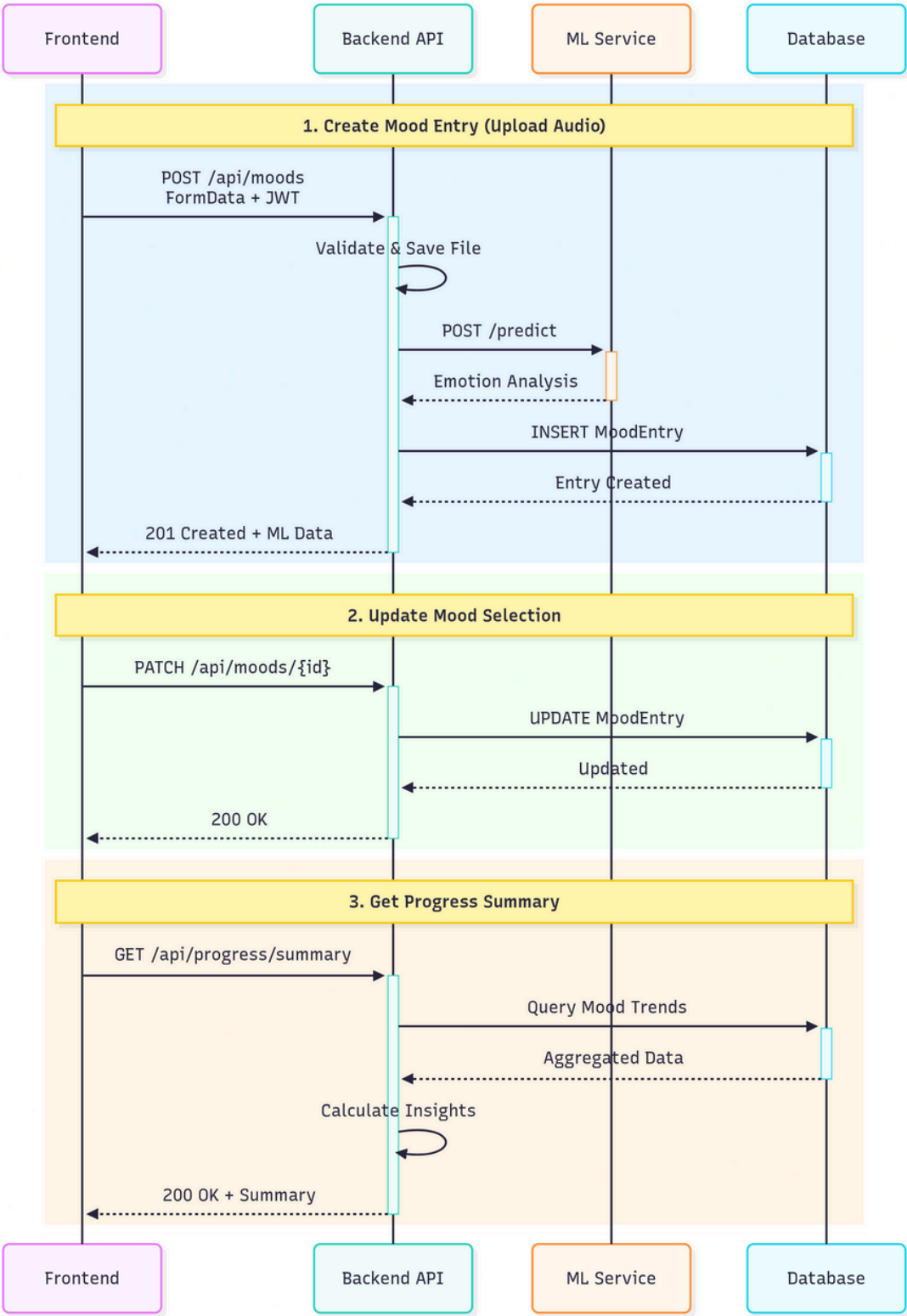




# Database Schema



# API Flow

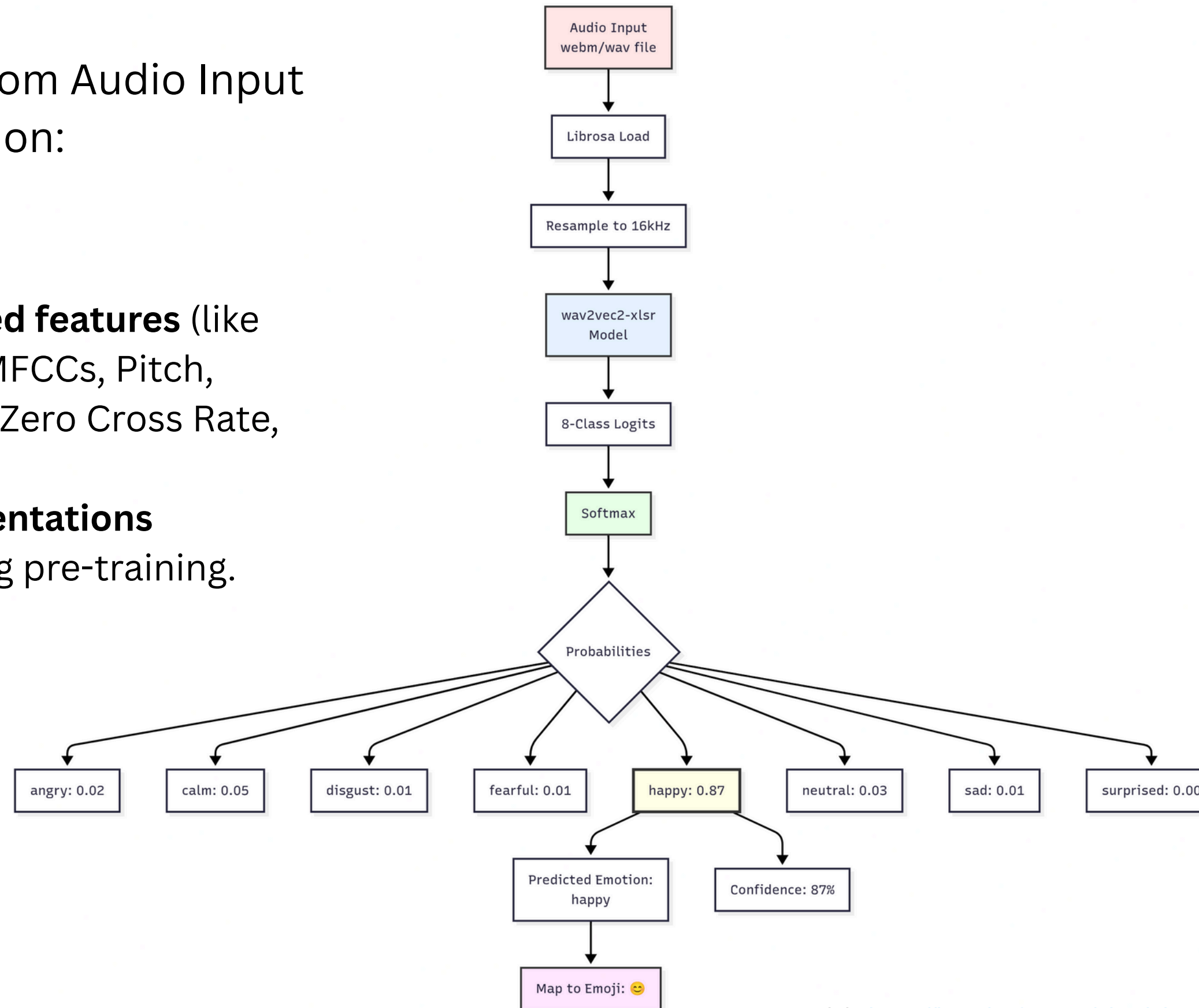




# ML Service Flow

Complete flow from Audio Input to Emoji Suggestion:

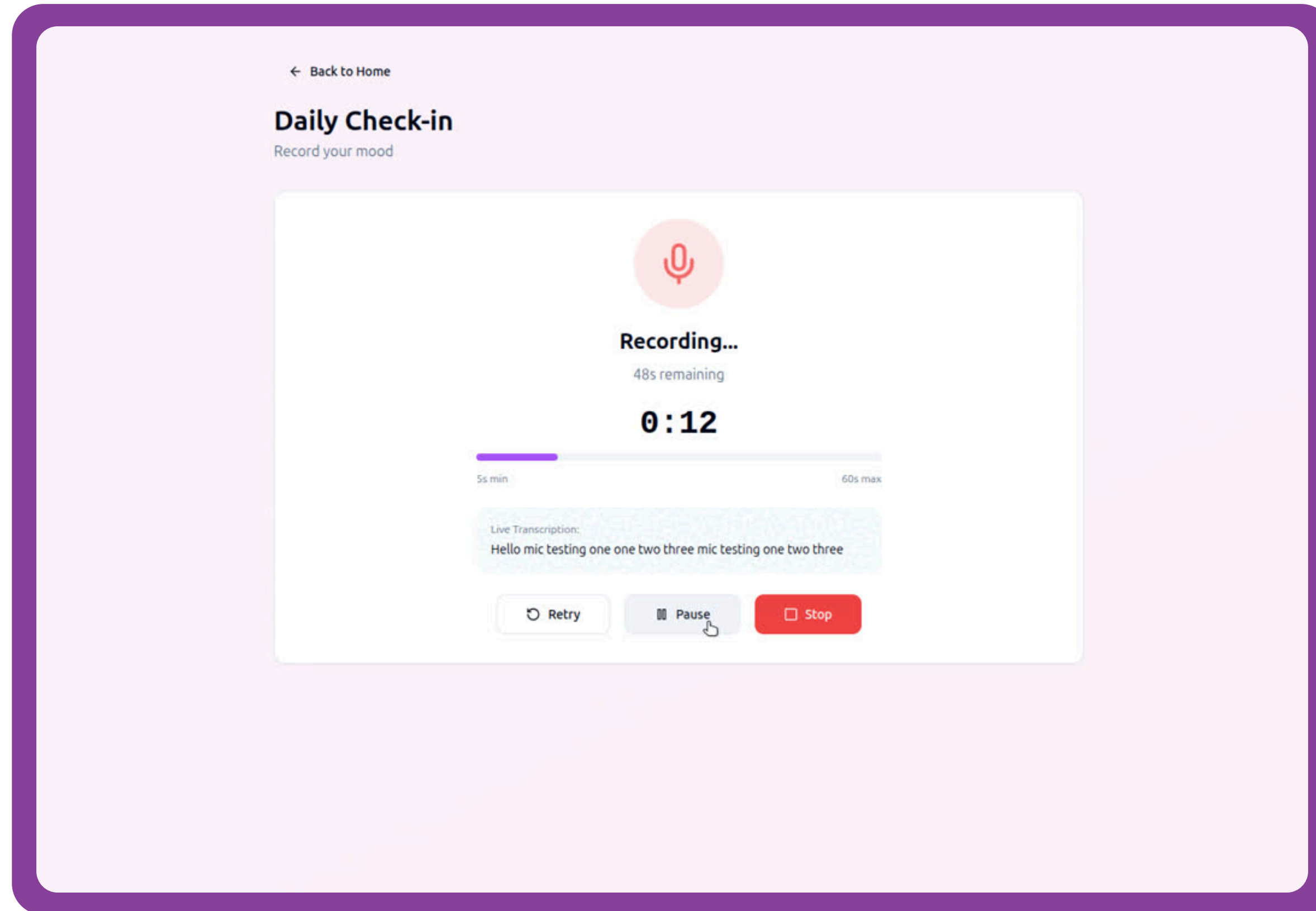
- **Not hand-crafted features** (like spectrograms, MFCCs, Pitch, Jitter, Shimmer, Zero Cross Rate, LPCC, etc.).
- **Learned representations** optimized during pre-training.



Link: <https://huggingface.co/ehcalabres/wav2vec2-lg-xlSr-en-speech-emotion-recognition>



# Demonstration



# Compromises in Prototyping

## Horizontal Compromises (UI Present, Backend Simplified)

| Feature                           | Our Implementation        | Compromises                                                      |
|-----------------------------------|---------------------------|------------------------------------------------------------------|
| Pattern Detection & Interventions | Alert popup displayed     | Generic placeholder text (no curated mental health resources)    |
| Multilingual Support              | Language selector present | English-only processing (model not fine-tuned on Hindi/Gujarati) |
| Privacy Controls                  | Audio Deletion available  | No local processing of audio                                     |

## Vertical Compromises

| Feature                   | Proposed Implementation                                                                    | Our Implementation                                              | Reason                                                                                                               |
|---------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Weekly Summary Feature    | Insights delivered once per week                                                           | Mood Analytics, No weekly summary.                              | Focused more on visualization for monthly and overall trends.                                                        |
| Emotion Recognition Model | Fine-tuned model on student speech data with feature analysis for mental health indicators | Pre-trained SOTA wav2vec2-lg-xlsr-en-speech-emotion-recognition | Lack of bandwidth (time) and ML expertise for dataset curation, feature engineering, and domain-specific fine-tuning |

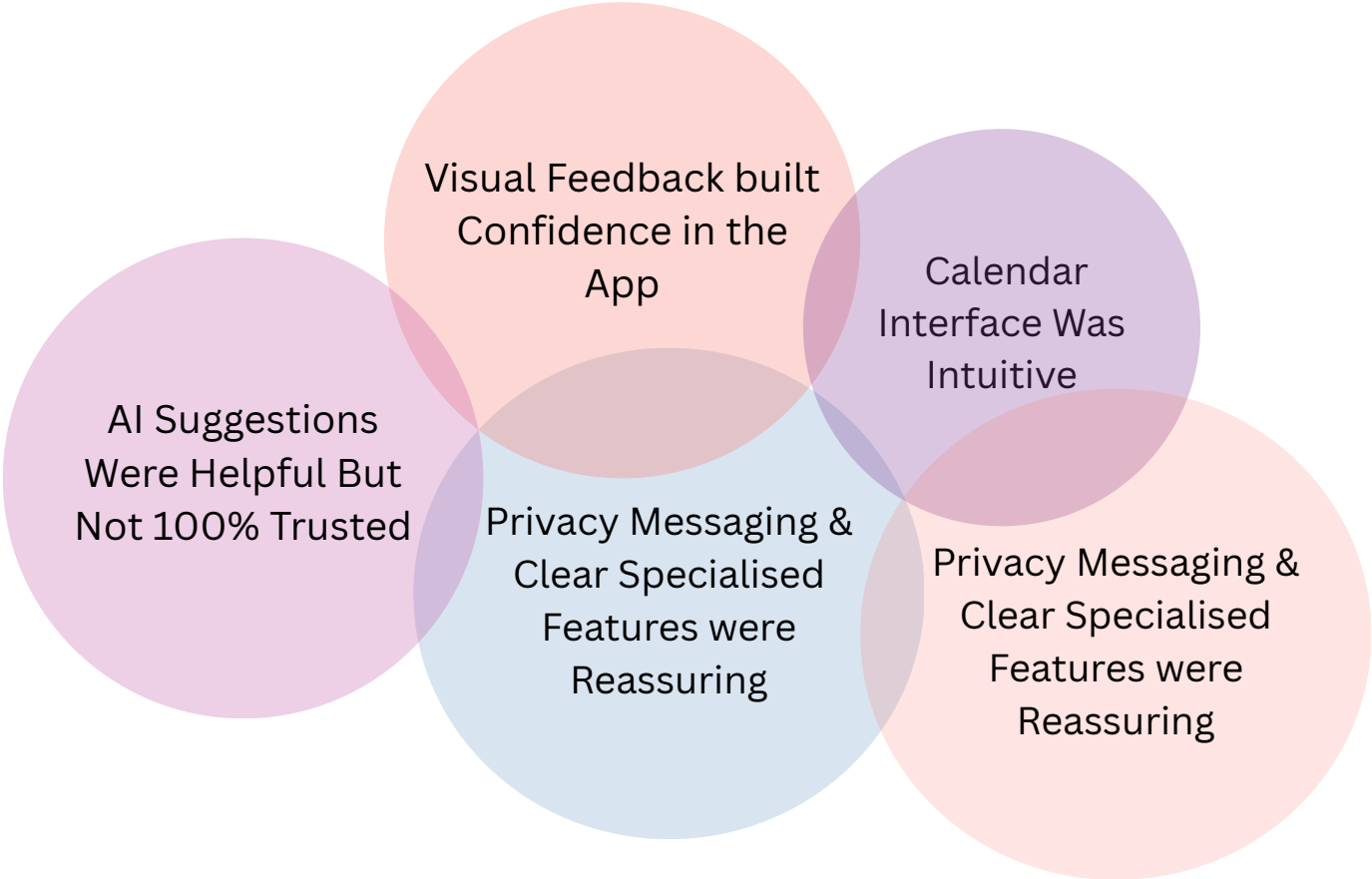
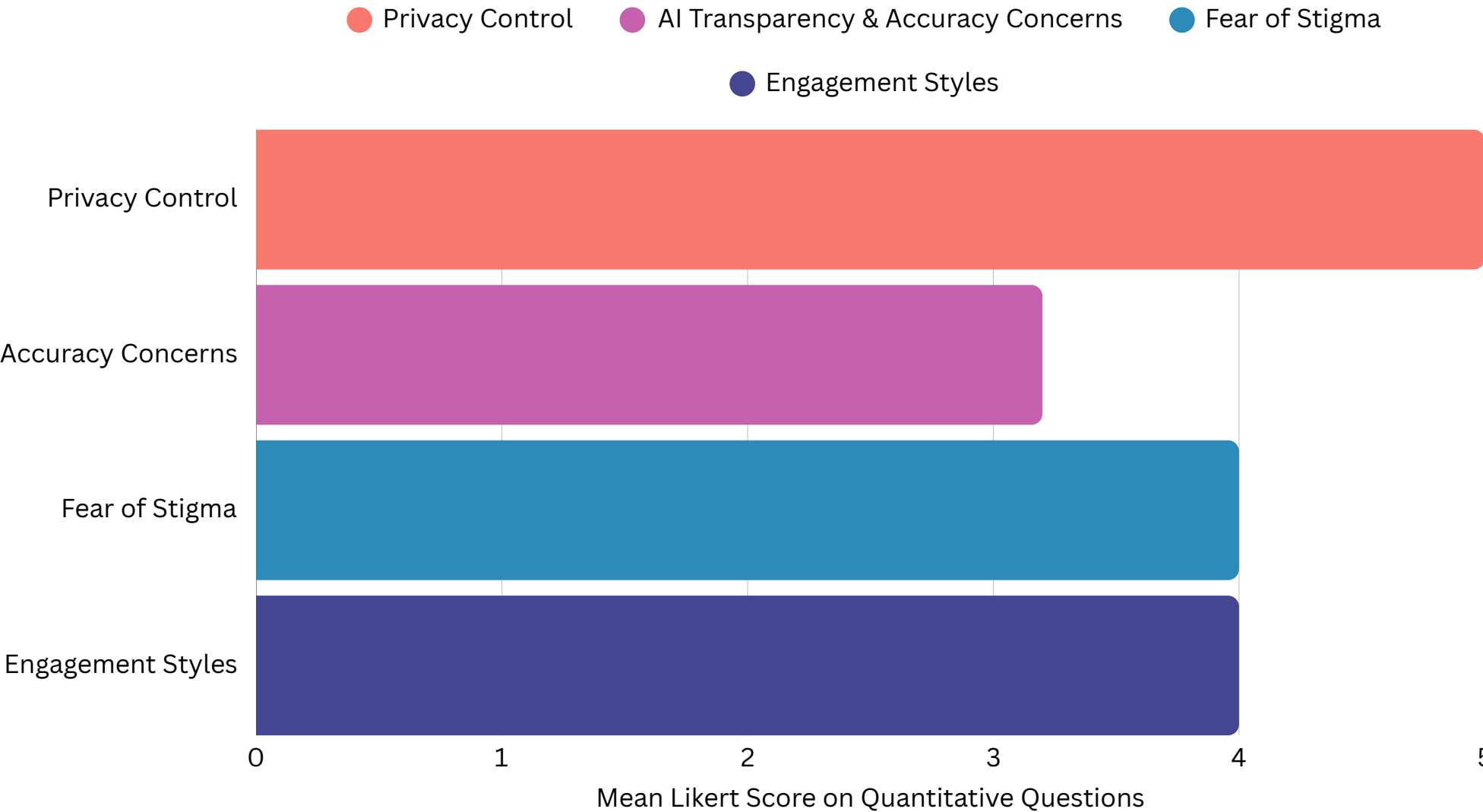
# User Evaluation

Mixed Methods Approach (n=5)

- System Usability Scale (SUS); Mean score: 85
- Basic Demographics
- Regarding Privacy & Data Control
- AI Transparency & Accuracy Concerns
- Fear of Stigma & Tone Sensitivity
- Personalization & Engagement Styles
- Additional Usability & Missing Elements

Quantitative analysis

Thematic analysis



# Conclusion

- Built a working speech-based mood monitoring system.
- Validated journaling metaphor as low-stigma and effective.
- Real ML integration revealed trust and explainability gaps.
- Identified challenges in long-term engagement.

